

Causal Fibres as an Evidence-Gated Hypothesis Ladder: A Hyperseed Formalization of Observability, Representation, Plasticity, Settlement, and Credit

ProtomegaTron / OpenClaw research formalization

23 July 2026

Abstract

The causal-fibres programme is best understood not as one architectural claim but as a sequence of defeasible hypotheses. This note formalizes its H0–H6 ladder as evidence-gated knowledge formation. A frozen read/write interface must first expose a teacher correction; rival representation classes must then compete at matched budget; selective plasticity and its support estimator must survive held-out tests; only afterward may predictive-coding settlement and a learned credit highway claim value. We define loss- and accuracy-based teacher-gap closure, factor-specific counterfactual closure, representation objects, support and drift measures, settlement modes, the amortization gap, and quantitative kill gates. The Hyperseed interpretation is methodological: freezing is a conservative falsifier, the representation contest is pluralistic hypothesis evaluation, and a symbolic cap is registration into an AtomSpace rather than post hoc labeling. Proof sketches show why energy descent does not identify causal credit, why teacher-clamped settlement is not deployment performance, and why orthogonal fibres are sufficient under common reducing subspaces but are not a universal model of neural representation.

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1 Scope, provenance, and epistemic status

The primary sources are the `causal-fibres` 0.4.0 release README, `TRANSFORMER_GPU_EXPERIMENT.md`, `REPRESENTATION_CONTEST.md`, and `CRITIQUE_DRIVEN_REVISION.md` [1]. The intended first GPU programme uses a six-layer synthetic-grammar student, a GPT-2-small teacher, and controlled counterfactuals for subject number, object number, tense, negation, and agreement. This note formalizes that programme; it does not report that its GPU gates have passed.

The conceptual novelty is the ordering of claims by what their evidence can actually establish. Earlier toy results established conditional mechanics, including exact preservation under supplied local support, a failure in which energy fell without correct causal credit (R9), a teacher-clamped/free deployment gap, underidentification of repeated modules, and recovery of exact blocks when a common block structure truly exists. They did not establish that realistic Transformer states possess a small orthogonal causal decomposition. We therefore use the following statuses: *defined*, *conditionally established*, *empirically passed*, *killed*, and *open*. In this paper H0–H6 are open empirical hypotheses.

2 Experimental objects and evidence records

Definition 2.1 (Frozen interface). *Let a base network with fixed parameters θ_0 expose reads*

$$h(x) = (h_{\ell_1}(x), \dots, h_{\ell_s}(x)) \in \mathcal{H}$$

at selected sites and accept a residual write $r(h(x))$ at specified logits or hidden states. The pair $I = (\text{read}, \text{write})$ is a frozen interface when θ_0 is held constant while the residual system is trained. A controlled co-adaptive interface I_ρ may later permit rank- ρ base updates inside a stated trust region.

Definition 2.2 (Matched budget). *For a representation arm R , define its budget vector*

$$B(R) = (P_{\text{ever}}, P_{\text{deploy}}, a, w, Q, t_{\text{train}}, t_{\text{deploy}}, \text{rank } r),$$

where the entries are parameters ever trained, deployment parameters, mean active features, code width, group count, and wall-clock and output-rank quantities. Arms are matched when designated coordinates of B are equal within preregistered tolerances; otherwise all coordinates and a capacity-normalized comparison must be reported.

Definition 2.3 (Evidence record). *An evidence record for hypothesis H_i is*

$$e_i = (D_{\text{train}}, D_{\text{val}}, D_{\text{held}}, I, R, B, \sigma, M, C),$$

where σ fixes seeds and software/hardware provenance, M is the vector of observed metrics, and C identifies controls. An admission predicate $G_i(e_i) \in \{0, 1\}$ states whether the preregistered gate passed. Advancement to H_{i+1} requires $G_i(e_i) = 1$; a result from a later mechanism cannot retroactively supply missing evidence for an earlier gate.

3 Teacher-gap closure and factor-specific correction

Let L_0, L_R, L_T be held-out losses for the frozen base, residual-augmented model, and teacher, respectively, evaluated on the same examples. Let A_0, A_R, A_T denote the corresponding accuracies.

Definition 3.1 (Teacher-gap closure). *When $L_0 > L_T$, loss closure is*

$$\Gamma_L(R; I) = \frac{L_0 - L_R}{L_0 - L_T}.$$

When $A_T > A_0$, accuracy closure is

$$\Gamma_A(R; I) = \frac{A_R - A_0}{A_T - A_0}.$$

A value 0 means no closure, 1 matches the teacher gap, values below zero degrade the base, and values above one surpass the teacher on the stated metric. If the denominator is nonpositive or statistically indistinguishable from zero, the corresponding closure is undefined and cannot pass a gate.

Ratios can be unstable near a vanishing denominator. Every report must therefore include raw losses or accuracies, denominators, uncertainty intervals, and per-example paired comparisons in addition to Γ .

Definition 3.2 (Counterfactual factor closure). *For causal factor j , let $\tau_j x$ edit only factor j by construction, and let $\delta_j^M(x) = f_M(\tau_j x) - f_M(x)$ be the model's output edit (for example, a logit vector). Define the teacher-referenced edit loss*

$$E_j(M) = \mathbb{E}_x \left[d(\delta_j^M(x), \delta_j^T(x)) \right].$$

Provided $E_j(0) > 0$, factor-specific closure is

$$\Gamma_j = 1 - \frac{E_j(R)}{E_j(0)}.$$

Selectivity is recorded by the off-factor spill

$$S_{j \rightarrow -j} = \mathbb{E}_x \sum_{k \neq j} d_k(\delta_j^R(x), \delta_j^T(x)),$$

or by a preregistered intervention-confusion matrix. Large total closure with poor Γ_j or high spill is capacity evidence, not factor alignment.

Kill criterion 3.3 (H0 interface gate). *For an ordinary backpropagation-trained linear, low-rank, or MLP residual on the exact frozen reads and writes,*

$$\max\{\Gamma_L, \Gamma_A\} > 0.50$$

on held-out data, with a confidence interval excluding the preregistered failure region. If no reasonable BP capacity curve passes, kill or redesign the interface before interpreting failures of fibres, PC, or credit routing.

4 Five rival representation classes

Let $h \in \mathbb{R}^d$ be the frozen read and let r_b be the residual write produced under representation class b .

Definition 4.1 (Dense residual). *A dense bottleneck uses an encoder $E : \mathbb{R}^d \rightarrow \mathbb{R}^k$ and write matrix $W : \mathbb{R}^k \rightarrow \mathbb{R}^o$:*

$$z = E(h), \quad r_b = Wz.$$

It is a raw-capacity control and makes no modularity claim.

Definition 4.2 (Sparse autoencoder). *An SAE has an overcomplete dictionary $D \in \mathbb{R}^{d \times m}$, $m > d$, with*

$$h \approx Dz, \quad z \in \mathbb{R}^m, \quad \|z\|_0 \ll m,$$

and a steering map $r_b = Wz$. Sparsity may be imposed by top-k, thresholding, or a stated penalty. Reconstruction alone does not establish causal meaning.

Definition 4.3 (Grouped sparse dictionary). *Let $\{S_q\}_{q=1}^Q$ be subsets of $\{1, \dots, m\}$, not necessarily disjoint. The code is group sparse under, for example,*

$$\Omega(z) = \sum_{q=1}^Q \lambda_q \|z_{S_q}\|_2, \quad h \approx Dz.$$

Overlaps permit a dictionary atom to participate in more than one candidate module and hence model superposition without forcing orthogonality.

Definition 4.4 (Orthogonal fibres). *An orthogonal fibre model posits*

$$\mathcal{H} = U_1 \oplus \dots \oplus U_Q, \quad U_p \perp U_q \ (p \neq q),$$

with orthogonal projections P_q , code $z_q = P_q h$, and $r_b = \sum_q W_q z_q$. A causal interpretation additionally requires selective interventions and identification; a low off-block operator loss alone is only structural evidence.

Definition 4.5 (Contextual fibre bundle). *Let $c \in \mathcal{C}$ index context, $z_{\text{canonical}}$ be a shared coordinate, and T_c an invertible or controlled low-rank local transport:*

$$z_{\text{local}} = T_c z_{\text{canonical}}, \quad r_b = W_c z_{\text{local}}.$$

Transition maps $T_{c'} = T_c T_c^{-1}$ relate charts. This class retains a canonical intervention or symbolic address while allowing context-dependent rotations.

Kill criterion 4.6 (H1 representation gate). *At matched budget, a structured arm must either (i) strictly improve a preregistered primary metric relative to both dense and SAE-plus-steering controls, with multiplicity-aware uncertainty, or (ii) add complementary value on intervention selectivity, held-out combinations, or continual drift while remaining noninferior on task performance. Otherwise the structured claim is killed, even if reconstruction is good.*

5 The H0–H6 ladder

Definition 5.1 (H0: frozen interface observability). *There exists a reasonable BP residual class on frozen interface I that closes more than half the teacher gap, including preregistered factor corrections. H0 asks whether the correction is visible and writable at all; it does not ask which structured ontology is correct.*

Definition 5.2 (H1: representation contest). *Among matched-budget rivals, at least one structured representation—SAE, grouped sparse dictionary, orthogonal fibres, or contextual bundle—beats or complements dense and SAE controls on correction plus an explicitly structural criterion. H1 may select different winners in different regimes.*

Definition 5.3 (H2: selective persistent plasticity). *Let parameters be partitioned into candidate supports $\Theta = \bigcup_q \Theta_q$. For task/context c , let $s_c \in [0, 1]^Q$ be supplied or learned responsibility and update*

$$\Delta\theta_q = -\eta s_{c,q} \nabla_{\theta_q} L_c.$$

For true support S_c^ , define off-support update mass*

$$D_{\text{off}}(c) = \frac{\sum_{q \notin S_c^*} \|\Delta\theta_q\|_2}{\sum_q \|\Delta\theta_q\|_2 + \epsilon},$$

and behavioral drift on unrelated probes

$$K_{\text{unrelated}}(c) = \mathbb{E}_{x \sim D_{-c}} \text{KL}(p_{\theta-}(\cdot|x) \parallel p_{\theta+}(\cdot|x)).$$

H2 holds only if gated exact-gradient plasticity reduces preregistered drift relative to ungated exact gradient, replay, and parameter-isolation controls while matching current-task performance within tolerance.

Definition 5.4 (H3: held-out generalization). *Train a router or support estimator on contexts $\mathcal{C}_{\text{train}}$ and evaluate on disjoint contexts or compositions $\mathcal{C}_{\text{held}}$. With thresholded prediction $\hat{S}_c$,*

$$P = \frac{|\hat{S}_c \cap S_c^*|}{|\hat{S}_c|}, \quad R = \frac{|\hat{S}_c \cap S_c^*|}{|S_c^*|}, \quad \text{F1}_c = \frac{2PR}{P+R}.$$

H3 requires mean held-out support $\text{F1} > 0.70$ and generalization of intervention fingerprints

$$\phi_{qj}(c) = d_j\left(f(x; c), f(\text{do}(z_q \leftarrow z'_q); c)\right)$$

to unseen contexts, not merely in-distribution support recovery.

Kill criterion 5.5 (H3 prerequisite gate). *If held-out support $\text{F1} \leq 0.70$, do not advance to PC/ePC or learned highways. A supplied oracle mask may still test conditional mechanics but must be labeled oracle and cannot count as learned generalization.*

Definition 5.6 (H4: PC/ePC settlement value). *Let g_{ff} be a feed-forward predictor, g_{rec} a matched recurrent control, and S_K a K -step PC/ePC settlement operator. At matched deployment compute and parameters, $H4$ holds only if S_K improves a preregistered quantity—sample efficiency, robustness, retention, representation quality, or loss—beyond both controls. Energy reduction by itself is not such a quantity.*

Definition 5.7 (Settlement modes and amortization gap). *For a common test distribution, distinguish*

$$\begin{aligned} y_{\text{pred}} &= g_{\text{ff}}(h), & y_{\text{free}} &= S_K(h; \emptyset), \\ y_{\text{con}} &= S_K(h; \xi_{\text{test}}), & y_{\text{clamp}} &= S_K(h; y_T), \end{aligned}$$

where ξ_{test} is admissible retrieval, memory, recurrence, or symbolic constraint information available at deployment, while y_T is an oracle teacher signal. For a loss $\mathcal{L}$, define

$$\Delta_{\text{amort}} = L_{\text{free}} - L_{\text{clamp}}, \quad L_m = \mathbb{E} \mathcal{L}(y_m, y^*).$$

One may also report the normalized gap $\tilde{\Delta}_{\text{amort}} = (L_{\text{free}} - L_{\text{clamp}}) / (L_{\text{pred}} - L_{\text{clamp}})$ when its denominator is positive.

Kill criterion 5.8 (H4 deployment kill metric). *Predictor-only, free, constrained, and teacher-clamped results must be reported separately. A large positive Δ_{amort} with no matched-compute free-settlement improvement kills a deployment claim for PC, even if clamped settlement is excellent. The clamped result remains an oracle diagnostic.*

Definition 5.9 (H5: constrained settlement exploitation). *H5 holds when admissible ξ_{test} contains genuinely new test-time information and constrained settlement exploits it better than controls given the same information and compute:*

$$L_{\text{con}} < \min(L_{\text{free}}, L_{\text{ff}+\xi}, L_{\text{rec}+\xi}) - \delta.$$

The information source, access policy, latency, and control exposure must be recorded. Repackaging training-time teacher labels as a “constraint” does not satisfy H5.

Definition 5.10 (H6: learned causal highway). *Let $g_{\text{exact},c}$ be exact autograd credit after responsibility gating and $\hat{g}_{\psi,c}$ a learned credit highway. Define directional error*

$$e_{\text{dir}} = 1 - \frac{\langle \hat{g}_{\psi,c}, g_{\text{exact},c} \rangle}{\|\hat{g}_{\psi,c}\| \|g_{\text{exact},c}\| + \epsilon},$$

local communication cost C_{comm} , update cost C_{upd} , and held-out post-update loss $L_{\text{post}}^{\text{held}}$. H6 holds only if the highway improves at least one preregistered target—locality, novel-context transfer, or total cost—while remaining noninferior in task and drift metrics to exact gated credit. Beating dense feedback alone is insufficient.

6 Proof sketches and limits

Proposition 6.1 (Energy descent does not identify causal credit). *A state or parameter update can decrease a scalar energy and improve the current task while assigning credit to the wrong causal support.*

Proof sketch. Take two coordinates (u, v) and task output $y = u + v$ with energy $E(u, v) = \frac{1}{2}(u + v - 1)^2$. Suppose the intervention semantics designate u as the only cause that should persistently change, while v controls an unrelated capability. At $(0, 0)$, updates $(\eta, 0)$ and $(0, \eta)$ for $0 < \eta < 2$ yield exactly the same lower energy $\frac{1}{2}(\eta - 1)^2$. The second update has zero overlap with the true support and may damage the unrelated capability, yet energy and current-task loss cannot distinguish it. More generally, scalar descent constrains the inner product with a gradient, not causal identity in a nonidentifiable parameter class. The R9 failure is therefore structurally possible: energy descent is evidence for optimization, not for correct causal credit. $\square$

Proposition 6.2 (Teacher-clamped settlement is not deployment performance). *Without a deployment-time channel carrying the teacher constraint, clamped settlement estimates an oracle conditional risk rather than the free deployment risk.*

Proof sketch. Let H be the frozen features and Y the desired output. A free estimator is measurable with respect to $\sigma(H)$; a clamped estimator is measurable with respect to $\sigma(H, Y_T)$, a strictly richer information set whenever $I(Y; Y_T | H) > 0$. By the value-of-information inequality, optimal risk under the richer sigma-algebra cannot exceed optimal risk under the poorer one. Thus low L_{clamp} may result entirely from oracle information. Unless Y_T or an equivalent constraint is available at deployment, it says nothing about attainable L_{free} . The observable difference Δ_{amort} measures the part the deployable inference process failed to amortize; it is not repaired by renaming the oracle mode. $\square$

Proposition 6.3 (When orthogonal fibres suffice). *For a finite-dimensional real inner-product space $\mathcal{H}$ and an operator family $\mathcal{A} = \{A_c\}$, an orthogonal decomposition $\mathcal{H} = \bigoplus_q U_q$ is a common block decomposition exactly when its orthogonal projections satisfy*

$$P_q A_c = A_c P_q \quad \text{for all } q, c.$$

For self-adjoint operators, common invariant subspaces are reducing and this condition yields orthogonal fibres.

Proof sketch. If P_q commutes with every A_c , then $A_c U_q \subseteq U_q$, so all off-block maps $P_p A_c P_q$ vanish for $p \neq q$. Conversely, if every A_c preserves every U_q in the orthogonal direct sum, its block matrix has no off-block entries and commutes with each P_q . For self-adjoint A_c , invariance of U_q implies invariance of $U_q^\perp$, giving a reducing subspace. $\square$

Remark 6.4 (Sufficient, not universal). *The proposition is a conditional sufficiency result, not a representation theorem for arbitrary neural states. Nonnormal or context-varying operators may lack non-trivial common reducing subspaces; noncommuting self-adjoint operators may generate an irreducible algebra; repeated isomorphic blocks can make the decomposition nonidentifiable; and superposed features may be better modeled by an overcomplete dictionary. Context-dependent conjugacies $A_c = T_c A_\star T_c^{-1}$ may instead support a bundle whose local charts rotate. Orthogonal fibres should therefore win a contest, not be assumed.*

7 Hyperseed interpretation

7.1 The ladder as evidence-gated knowledge formation

Let K_i be the accepted knowledge state after stage i . The update is

$$K_{i+1} = \begin{cases} K_i \cup \{\langle H_i, e_i, \text{supported} \rangle\}, & G_i(e_i) = 1, \\ K_i \cup \{\langle H_i, e_i, \text{killed} \rangle\}, & G_i(e_i) = 0. \end{cases}$$

This is a Hyperseed-style formation rule: evidence is not decoration attached to a favored ontology, but a gate controlling which concepts may acquire stronger status. The chain is not logically monotone in the sense that H_1 implies H_0 ; rather, it is procedurally conservative: evidence generated under an unpassed prerequisite cannot license the downstream interpretation. Alternative surviving representations may coexist with scoped truth values.

7.2 Freezing as conservative falsifier

Freezing isolates the proposition “the correction is readable and writable through I ” from the much larger proposition “the whole substrate can adapt until the task works.” If a frozen BP residual fails H_0 , adding PC or a highway only introduces more degrees of freedom and weakens diagnosis. This implements a Hyperseed conservativity principle: extend the ontology or dynamics only after the least expressive admissible account fails under a recorded test. Freezing is not dogma; after its result is registered, a rank-limited I_ρ with anchor drift constraints becomes a new, explicitly different hypothesis.

7.3 Pluralistic representation evaluation

The representation contest instantiates pluralistic hypothesis evaluation. Each class supplies a different ontology of the same activation: dense latent capacity, sparse superposed atoms, overlapping functional groups, orthogonal subspaces, or transported contextual coordinates. Matched budgets make these alternatives commensurable without pretending they say the same thing. Hyperseed truth values should be indexed by dataset, intervention family, budget, and context; “orthogonal fibres supported” is not promoted to “all cognition decomposes orthogonally.”

7.4 Symbolic cap as registration

A symbolic cap is scientifically meaningful only when the mapping is registered before the decisive behavioral test and participates causally in it. Let α_q be a neural coordinate or group, a_q an AtomSpace

atom, and π provenance. A minimal MeTTa-style registration is conceptually

(CausalCoordinate a_q q),
 (Grounding a_q α_q π),
 (SupportStatus a_q intervention-aligned),
 (Evidence a_q e_i).

A rule-derived atom must then induce a selective neural intervention and the predicted behavioral consequence on hard negatives and held-out compositions. Naming a feature after inspecting successful outputs is post hoc interpretation; it is not registration. AtomSpace/MeTTa is valuable here because identity, provenance, evidence status, and rule-mediated effects can be explicit and queryable [3, 2].

8 Decision table and reporting contract

Gate	Claim	Required comparison	Kill consequence
H0	Frozen interface exposes correction	BP residual capacity curve; Γ_L or $\Gamma_A > 0.50$	Change reads/writes or test controlled substrate adaptation.
H1	Structured code adds value	Dense and SAE+steering at matched budget	Retain simpler winner; no fibre-emergence claim.
H2	Selective plasticity protects unrelated function	Ungated exact gradient, replay, isolation	Do not attribute retention to causal routing.
H3	Support and fingerprints generalize	Disjoint contexts/compositions; $F1 > 0.70$	No PC/highway escalation; oracle masks remain diagnostics.
H4	PC/ePC settlement adds value	Feed-forward and recurrent controls at matched compute	Energy descent or clamped success alone is insufficient.
H5	Constraints add deployable information	All controls receive the same ξ_{test}	Treat benefit as oracle leakage or ordinary control capacity.
H6	Learned highway beats exact gated credit	Exact autograd on locality, transfer, cost, task, drift	Use exact gated credit; do not promote the highway.

Every report must include dataset splits, counterfactual generator, frozen and co-adaptive interfaces, all budget coordinates, raw and normalized task metrics, factor closure and spill, support precision/recall/F1, off-support update mass, unrelated KL drift, finite curriculum-order distances, settlement modes, Δ_{amort} , wall clock, seeds, software commit, and uncertainty. Negative results remain first-class registered evidence.

9 Conjectures and open questions

Conjecture 9.1 (Contextual rescue). *When a global orthogonal contest fails but context operators are approximately conjugate to a stable canonical family, a contextual bundle will improve held-out intervention fingerprint transport relative to both a single orthogonal basis and an unconstrained dense adapter at matched budget.*

Conjecture 9.2 (Selective-plasticity value). *Conditional on H1 and accurate support, responsibility-gated exact gradients will reduce finite curriculum-order sensitivity and unrelated KL drift without reducing adaptation on the current task. Failure would show that the selected representation is not a useful persistent-plasticity unit even if it predicts teacher residuals.*

Open questions include how uncertainty in teacher quality should propagate through Γ ; whether support F1 is calibrated to the downstream cost of false positives versus false negatives; when a bundle's

transports are identifiable rather than gauge choices; and what admissible information policy distinguishes symbolic constraint from benchmark leakage. These questions reinforce the central discipline: later machinery must earn its interpretation through earlier, scoped evidence.

10 Conclusion

The causal-fibres programme becomes sharper when treated as a ladder of evidence admissions rather than a package of mutually supporting intuitions. H0 can kill an interface before structure is blamed. H1 can reject orthogonality without rejecting useful adaptation. H2 and H3 distinguish conditional preservation from learned generalization. H4 and H5 separate optimization dynamics from deployable information. H6 makes a learned highway compete with the conservative exact-credit baseline. In Hyperseed terms, the important object is not “the fibre” alone but the fibre together with its scope, rivals, falsifier, evidence provenance, and permitted inferences.

References

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